TextHack: High-Performance Pattern Matching & Duplicate Document Detection

B.Tech Final Year Project Review | Department of AI & Data Science, KL University

Presenter Name: Valakonda Pavan Kowshik

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Problem Statement & Objectives

CHALLENGES

Challenges: Manual plagiarism verification bottlenecks and legacy O(N*M) string search inefficiencies.

OBJECTIVES

Objectives: Implement linear-time O(N + M) pattern search and constant-time O(1) duplicate document registration.

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Literature Review & Existing Systems

CONTEXT

Context

Limitations of legacy tools like Turnitin (high memory overhead and heavy database indexing).

RESOLUTION

Resolution

Combining in-memory HashSets with KMP prefix table optimizations.

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Proposed System Architecture

WORKFLOW

Workflow

Stream-based file text ingestion, normalization, processing engine, and registry layer.

DESIGN

Design

Modular layout separating string pattern matching from duplicate lookup audit trails.

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Methodology & Algorithms

01

KMP Algorithm

Utilizes LPS prefix tables to skip redundant character checks in O(N + M) time.

02

HashSet Registry

Stores normalized text signatures in JVM memory for instant O(1) duplicate rejection.

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Tools & Technologies



Core Stack

Java, JavaScript, HTML5, and Tailwind CSS.



Environment

VS Code development workspace with interactive UI components.

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Implementation & Results

< 0.1ms

Performance: Sub-millisecond execution latency (< 0.1ms) for duplicate lookups.

REAL-TIME UI

Output: Real-time UI logs showing successful unique entries vs. instant duplicate alerts.

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Advantages & Real-World Applications



Advantages

Zero speed degradation as corpus scales, constant-time performance, and high accuracy.



Applications

Academic assignment screening, code repository redundancy checks, and enterprise text analytics.

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Challenges & Future Enhancements

CHALLENGES

Challenges

Managing memory allocation during large-scale text parsing.

FUTURE SCOPE

Future Scope

Integrating TF-IDF and Cosine Similarity for advanced semantic fuzzy matching.

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CONCLUSION

Conclusion & Q&A



Summary: Successfully designed and deployed a high-performance text analytics and duplicate prevention framework.

Closing: Thank you. Open for questions and discussion.

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